

PART A — QUESTIONS

AF23-001

Question:

Solve: $25x - 18 = 132$

- A) 5
- B) 6
- C) 7
- D) 8

AF23-002

Question:

If $23n + 12 = 173$, what is n ?

- A) 5
- B) 6
- C) 7
- D) 8

AF23-003

Question:

Simplify: $10(3x - 2) - 11x$

- A) $19x - 20$
- B) $30x - 20$
- C) $19x - 2$
- D) $41x - 20$

AF23-004

Question:

A company charges \$52 per batch plus an \$18 processing fee. Which expression represents the total charge for b batches?

- A) $52b + 18$
- B) $18b + 52$
- C) $70b$
- D) $52 - 18b$

AF23-005

Question:

If $f(x) = 20x - 67$, what is $f(8)$?

- A) 83
- B) 93
- C) 160
- D) 227

AF23-006

Question:

Which ordered pair satisfies the equation $y = 18x + 5$?

- A) (1, 22)
- B) (2, 40)
- C) (3, 59)
- D) (4, 76)

AF23-007

Question:

Solve: $28(x - 7) = 196$

- A) 11
- B) 12
- C) 13
- D) 14

AF23-008

Question:

A pattern begins: 16, 38, 60, 82, ... What is the next number?

- A) 102
- B) 103
- C) 104
- D) 105

AF23-009

Question:

Solve: $x/55 = 3$

- A) 58
- B) 110
- C) 165
- D) 220

AF23-010

Question:

A line has a slope of -20 and crosses the y-axis at 17. Which equation represents the line?

- A) $y = 17x - 20$
- B) $y = -20x + 17$
- C) $y = 20x + 17$
- D) $y = -17x + 20$

AF23-011

Question:

If 6 pallets contain 252 parts, how many parts are in 5 pallets at the same rate?

- A) 180
- B) 200
- C) 210
- D) 220

AF23-012

Question:

Solve: $420 - 38x = 268$

- A) 3
- B) 4
- C) 5
- D) 6

AF23-013

Question:

Which expression is equivalent to $132v - 61v - 22$?

- A) $71v - 22$
- B) $193v - 22$
- C) $71v + 22$
- D) $132v - 83$

AF23-014

Question:

A storage area starts with 2,400 units. Workers add 200 units each week. Which equation gives the total number of units U after w weeks?

- A) $U = 2,400w + 200$
- B) $U = 200w + 2,400$
- C) $U = 2,600w$
- D) $U = 2,400 - 200w$

AF23-015

Question:

If $y = -30x + 500$, what is y when $x = 14$?

- A) 70
- B) 80
- C) 420
- D) 920

AF23-016

Question:

Solve: $40x - 96 = 28x + 60$

- A) 11
- B) 12
- C) 13
- D) 14

AF23-017

Question:

Which value of x makes $24x - 10$ less than 302?

- A) 12
- B) 13
- C) 14
- D) 15

AF23-018

Question:

A table shows a linear relationship.

x : 0, 1, 2, 3

y : -12, 11, 34, 57

What is the rule?

- A) $y = 23x - 12$
- B) $y = 12x + 23$
- C) $y = -12x + 23$
- D) $y = 23x + 12$

AF23-019

Question:

Solve: $24(x + 5) = 21x + 159$

- A) 11
- B) 12
- C) 13
- D) 14

AF23-020

Question:

If $a = -17$ and $b = 14$, what is $5a + 6b$?

- A) -169
- B) -1
- C) 1
- D) 169

AF23-021

Question:

Which point is located in Quadrant IV?

- A) (19, -10)
- B) (-19, 10)
- C) (-19, -10)
- D) (19, 10)

AF23-022

Question:

A worker checks 300 items in 15 minutes. At the same rate, how many items can the worker check in 6 minutes?

- A) 100
- B) 110
- C) 120
- D) 130

AF23-023

Question:

Solve for n : $H = 45n - 90$

- A) $n = H + 90$
- B) $n = (H + 90)/45$
- C) $n = 45(H + 90)$
- D) $n = H/45 - 90$

AF23-024

Question:

What is the slope of a line that goes through (11, 24) and (17, 120)?

- A) 14
- B) 15
- C) 16
- D) 96

AF23-025

Question:

Simplify: $100x - 39 - 89x + 58$

- A) $11x + 19$
- B) $189x + 19$
- C) $11x - 97$
- D) $189x - 97$

AF23-026

Question:

If $g(x) = x^2 - 13x$, what is $g(17)$?

- A) 51
- B) 68
- C) 221
- D) 289

AF23-027

Question:

A sequence follows the rule “subtract 9, then multiply by 6.” If the first term is 15, what is the second term?

- A) 6
- B) 24
- C) 36
- D) 90

AF23-028

Question:

Solve: $28(x - 6) + 42 = 266$

- A) 11
- B) 12
- C) 13
- D) 14

AF23-029

Question:

A line crosses the y-axis at 20 and falls 12 units for every 1 unit to the right. Which equation represents the line?

- A) $y = 12x + 20$
- B) $y = -12x + 20$
- C) $y = 20x - 12$
- D) $y = -20x + 12$

AF23-030

Question:

If $36/54 = x/162$, what is x ?

- A) 96
- B) 102
- C) 108
- D) 114

AF23-031

Question:

Solve: $-50x + 250 = -450$

- A) -14
- B) -10
- C) 10
- D) 14

AF23-032

Question:

A warehouse starts with 8,400 clips. Each assembly uses 420 clips. Which expression gives the number of clips left after a assemblies?

- A) $8,400 + 420a$
- B) $8,400 - 420a$
- C) $420a - 8,400$
- D) $8,400 \div 420a$

AF23-033

Question:

Which equation has $x = 32$ as its solution?

- A) $2x + 20 = 84$
- B) $5x - 50 = 105$
- C) $x/16 + 7 = 10$
- D) $3x - 18 = 80$

PART B — ANSWER KEY + EXPLANATIONS

AF23-001 — Correct: B — Explanation:

Solve $25x - 18 = 132$. Add 18 to both sides: $25x = 150$. Divide by 25: $x = 6$. The most common mistake is subtracting 18 instead of adding 18.

AF23-002 — Correct: C — Explanation:

Start with $23n + 12 = 173$. Subtract 12 from both sides: $23n = 161$. Divide by 23: $n = 7$. A common mistake is stopping at 161 and forgetting to divide by 23.

AF23-003 — Correct: A — Explanation:

Distribute first: $10(3x - 2) = 30x - 20$. Then subtract $11x$: $30x - 11x - 20 = 19x - 20$. A common mistake is forgetting to multiply -2 by 10.

AF23-004 — Correct: A — Explanation:

The company charges \$52 for each batch, so that part is $52b$. The \$18 processing fee is added one time. The total charge is $52b + 18$. A common mistake is treating the processing fee as the per-batch charge.

AF23-005 — Correct: B — Explanation:

Substitute 8 for x : $f(8) = 20(8) - 67 = 160 - 67 = 93$. A common mistake is adding 67 instead of subtracting it.

AF23-006 — Correct: C — Explanation:

Test the choices in $y = 18x + 5$. For $(3, 59)$, $18(3) + 5 = 54 + 5 = 59$, which matches the y -value. A common mistake is choosing a point that is close but does not make the equation true.

AF23-007 — Correct: D — Explanation:

Solve $28(x - 7) = 196$. Divide both sides by 28: $x - 7 = 7$. Add 7: $x = 14$. A common mistake is adding 7 before undoing the multiplication by 28.

AF23-008 — Correct: C — Explanation:

The pattern increases by 22 each time: 16, 38, 60, 82. Add 22 to 82: 104. A common mistake is assuming the increase is 20 because the numbers are close to round numbers.

AF23-009 — Correct: C — Explanation:

$x/55 = 3$ means x divided by 55 equals 3. Multiply both sides by 55: $x = 165$. A common mistake is adding 55 and 3 instead of multiplying.

AF23-010 — Correct: B — Explanation:

Slope-intercept form is $y = mx + b$. The slope is -20 and the y-intercept is 17, so the equation is $y = -20x + 17$. A common mistake is switching the slope and y-intercept.

AF23-011 — Correct: C — Explanation:

Find the unit rate: $252 \text{ parts} \div 6 \text{ pallets} = 42 \text{ parts per pallet}$. For 5 pallets: $42 \times 5 = 210$. A common mistake is multiplying 252 by 5 without finding the rate first.

AF23-012 — Correct: B — Explanation:

Solve $420 - 38x = 268$. Subtract 420 from both sides: $-38x = -152$. Divide by -38: $x = 4$. A common mistake is losing the negative sign after subtracting 420.

AF23-013 — Correct: A — Explanation:

Combine like terms: $132v - 61v = 71v$. The constant -22 stays the same. The expression becomes $71v - 22$. A common mistake is combining the constant with the variable terms.

AF23-014 — Correct: B — Explanation:

The storage area starts with 2,400 units. Workers add 200 units each week, so after w weeks, $200w$ units are added. The total is $U = 200w + 2,400$. A common mistake is making 2,400 the weekly rate.

AF23-015 — Correct: B — Explanation:

Substitute $x = 14$ into $y = -30x + 500$: $y = -30(14) + 500 = -420 + 500 = 80$. A common mistake is treating $-30(14)$ as positive 420.

AF23-016 — Correct: C — Explanation:

Solve $40x - 96 = 28x + 60$. Subtract $28x$: $12x - 96 = 60$. Add 96: $12x = 156$. Divide by 12: $x = 13$. A common mistake is adding 96 to only one side.

AF23-017 — Correct: A — Explanation:

Solve $24x - 10 < 302$. Add 10: $24x < 312$. Divide by 24: $x < 13$. The only listed value less than 13 is 12. A common mistake is choosing 13, but $24(13) - 10 = 302$, not less than 302.

AF23-018 — Correct: A — Explanation:

The y-values increase by 23 each time x increases by 1, so the slope is 23. When $x = 0$, $y = -12$, so the y-intercept is -12. The rule is $y = 23x - 12$. A common mistake is using +12 instead of -12.

AF23-019 — Correct: C — Explanation:

Distribute: $24x + 120 = 21x + 159$. Subtract $21x$: $3x + 120 = 159$. Subtract 120: $3x = 39$. Divide by 3: $x = 13$. A common mistake is forgetting to multiply 5 by 24.

AF23-020 — Correct: B — Explanation:

Substitute $a = -17$ and $b = 14$: $5a + 6b = 5(-17) + 6(14) = -85 + 84 = -1$. A common mistake is treating $5(-17)$ as positive 85.

AF23-021 — Correct: A — Explanation:

Quadrant IV has a positive x-coordinate and a negative y-coordinate. The point $(19, -10)$ fits this description. A common mistake is choosing $(-19, 10)$, which is Quadrant II.

AF23-022 — Correct: C — Explanation:

300 items in 15 minutes means $300 \div 15 = 20$ items per minute. In 6 minutes: $20 \times 6 = 120$. A common mistake is multiplying 300 by 6 without finding the rate first.

AF23-023 — Correct: B — Explanation:

Start with $H = 45n - 90$. Add 90 to both sides: $H + 90 = 45n$. Divide by 45: $n = (H + 90)/45$. A common mistake is dividing H by 45 first and then adding 90.

AF23-024 — Correct: C — Explanation:

Slope equals change in y divided by change in x. From $(11, 24)$ to $(17, 120)$, y increases by 96 and x increases by 6. The slope is $96 \div 6 = 16$. A common mistake is using 96 as the slope without dividing by the change in x.

AF23-025 — Correct: A — Explanation:

Combine like terms: $100x - 89x = 11x$, and $-39 + 58 = 19$. The simplified expression is $11x + 19$. A common mistake is writing $11x - 19$ because the positive 58 is missed.

AF23-026 — Correct: B — Explanation:

Substitute 17 for x: $g(17) = 17^2 - 13(17) = 289 - 221 = 68$. A common mistake is calculating 17^2 as 34 instead of 289.

AF23-027 — Correct: C — Explanation:

Start with 15. Subtract 9: 6. Then multiply by 6: 36. A common mistake is multiplying first and then subtracting 9.

AF23-028 — Correct: D — Explanation:

Solve $28(x - 6) + 42 = 266$. Distribute: $28x - 168 + 42 = 266$. Combine: $28x - 126 = 266$. Add 126: $28x = 392$. Divide by 28: $x = 14$. A common mistake is forgetting to combine -168 and +42.

AF23-029 — Correct: B — Explanation:

“Falls 12 units for every 1 unit to the right” means the slope is -12. The line crosses the y-axis at 20, so the equation is $y = -12x + 20$. A common mistake is using a positive slope because the number 12 is given.

AF23-030 — Correct: C — Explanation:

Use the proportion $36/54 = x/162$. Since $54 \times 3 = 162$, multiply 36 by 3: $x = 108$. A common mistake is adding 108 to the numerator because 54 becomes 162.

AF23-031 — Correct: D — Explanation:

Solve $-50x + 250 = -450$. Subtract 250 from both sides: $-50x = -700$. Divide by -50: $x = 14$. A common mistake is giving -14 because both sides contain negative numbers.

AF23-032 — Correct: B — Explanation:

The warehouse starts with 8,400 clips. Each assembly uses 420 clips, so a assemblies use $420a$ clips. The number left is $8,400 - 420a$. A common mistake is adding $420a$ even though clips are being used.

AF23-033 — Correct: A — Explanation:

Test $x = 32$. In choice A, $2(32) + 20 = 64 + 20 = 84$, so the equation is true. A common mistake is choosing a nearby equation without checking both sides exactly.